From Econometrics to Machine Learning

MFE5340: AI in FE: Quantitative Investment

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- **Academia considers tradable portfolios as factors.**
- This portfolio is referred to as the *factor-mimicking portfolio* and the return of this portfolio represents the return of the factor.
- One of the most common methods for constructing factor-mimicking portfolios is the **portfolio sort** technique popularized by Fama and French.

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An Example

1 Portfolio Sort

- In portfolio sort, the variable used to rank stocks (such as BM) is called the *sorting variable*.
- **The core idea of the method is to use the stock's BM as a proxy for its exposure to the factor.**
- This method does not assume that the variable value equals factor exposure, nor does it assume any specific mathematical relationship between the two. The method simply assumes that the variable is correlated with the factor exposure.
- For example, with BM, the method assumes that stocks with high BM have higher exposure to the value factor, and those with low BM have lower exposure, and that's it.

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Procedure

1 Portfolio Sort

1. **Sorting:** Define the stock pool and rank all the stocks in the pool based on the sorting variable from high to low, or low to high, depending on the relationship between the sorting variable and stock return (i.e., positively or negatively correlated).

2. **Grouping:** Divide all stocks into L groups based on their ranking (by convention $L = 10$). A long/short hedged portfolio is then constructed by going long the stocks in the highest-ranked group and shorting the stocks in the lowest-ranked group. This portfolio is referred to as a *spread* portfolio.
   - The long and short legs are dollar neutral.
   - Value or equal weighting for long and short legs.

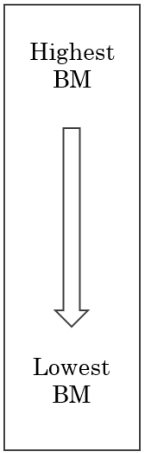
3. **Rebalance:** Since the value of the sorting variable for individual stocks changes over time, it is necessary to periodically repeat the above steps to rebalance the factor-mimicking portfolio.
   - Academic research often rebalances the portfolio monthly or even annually. But in industry practice, it is not uncommon to rebalance a factor portfolio daily.

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Procedure

1 Portfolio Sort

Step 1: Sort stocks



Step 2: Group stocks



Step 3: Form Portfolio



Long Position

Short Position

Step 4: Calculate Return

Portfolio Return = Return of Long Leg – Return of Short Leg

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Testing Factor's Risk Premium

1 Portfolio Sort

- Let $\{\lambda_t\}, t = 1, 2, \cdots, T$ represent the time series of factor returns, then the estimated risk premium $\hat{\lambda}$ and its standard error s.e.$(\hat{\lambda})$ are given by:

$$\hat{\lambda} = \frac{1}{T}\sum_{t=1}^{T}\lambda_t, \quad \text{s.e.}(\hat{\lambda}) = \frac{\text{std}(\lambda_t)}{\sqrt{T}},$$

  where std(λ_t) is the sample standard deviation of λ_t.

- With $\hat{\lambda}$ and s.e.$(\hat{\lambda})$, we can construct the t-statistic under the null hypothesis (i.e., $\lambda = 0$):

$$t\text{-statistic} = \frac{\hat{\lambda}}{\text{s.e.}(\hat{\lambda})},$$

  which follows a t-distribution with $T - 1$ degrees of freedom (why $T - 1$? because we adjust the degree of freedom when calculating the sample standard deviation!).

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- A good factor should be able to explain the cross-sectional variation in individual stock excess returns. $\Rightarrow$ Whether the returns of the L portfolios formed based on the sorting variable exhibit strong monotonicity.

- **Rank correlation coefficient** between the portfolio returns and the sorting variable groups. It measures the *monotonic relationship* between two random variables.

- Let X_r denote the ranks of portfolios' returns, and X_g note the sorting variable groups. Then, the rank correlation coefficient ρ_s is:

$$\rho_s = \frac{\text{cov}(X_r, X_g)}{\sigma_{X_r}\sigma_{X_g}}.$$

  The range of ρ_s is from -1.0 to 1.0.

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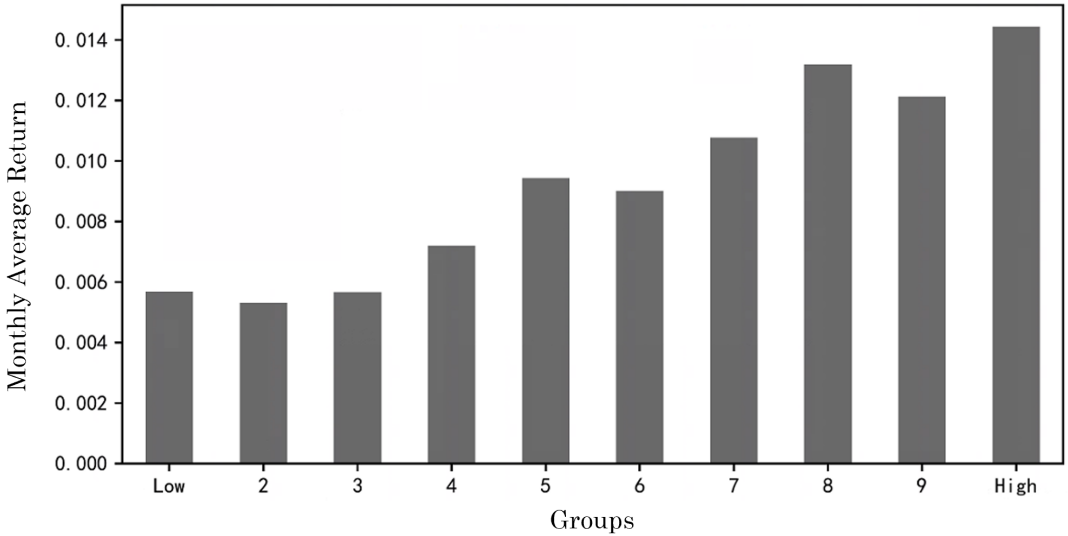
An Example: Book-to-Market Ratio

- At each month t (i.e. each cross-section), stocks in the Chinese stock market are divided into $L = 10$ groups (denoted as Low, 2, $\cdots$, 9, and High, where Low represents the group with the lowest BM values, and High represents the group with the highest BM values).
- Stocks in each group are weighted by market capitalization (i.e., value weighting), and the portfolio is rebalanced monthly.
- The factor's monthly average return is 0.88%, with a standard error of 0.47%, a t-statistic of 1.85, and a p-value of 0.07, indicating that we can reject the null hypothesis at the 0.1 significance level.

	Low	1	2	3	4	5	6	7	8	High	High−Low
Average Return (%)	0.57	0.53	0.57	0.72	0.94	0.90	1.08	1.32	1.21	1.44	0.88
Std Error (%)	0.57	0.56	0.55	0.58	0.56	0.53	0.57	0.58	0.56	0.58	0.47
t-statistic	1.00	0.95	1.03	1.24	1.69	1.69	1.90	2.27	2.15	2.48	1.85
p-value	0.32	0.35	0.30	0.22	0.09	0.09	0.06	0.02	0.03	0.01	0.07

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An Example: Book-to-Market Ratio



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Issue and Solution

- Previous portfolio sort is based on a single variable, and therefore it is also referred to as *univariate sorting*.
- Ideally, a factor-mimicking portfolio should have no exposure to other factors, but univariate sorting struggles to control the influence of other factors.
- For example, if all high-BM stocks are large-cap stocks and all low-BM stocks are small-cap stocks, then the factor portfolio will simultaneously go long large-cap stocks and short small-cap stocks.
- **Solution:** It is common to use more than one variable for double sorting or triple sorting, collectively known as multivariate sorting.

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Example: Profitability

1 Portfolio Sort

- The **profitability factor** reflects the tendency for stocks with higher profitability, often measured by metrics such as return on equity (ROE) or operating profits, to outperform those with lower profitability.

Table: Monthly Average Returns from Univariate ROE(TTM) Sorting (%)

Low	2	3	4	5	6	7	8	9	High	High−Low
Panel A: Equal-Weighted										
1.22	1.29	1.33	1.34	1.33	1.36	1.43	1.35	1.40	1.38	0.16
(1.59)	(1.68)	(1.75)	(1.81)	(1.84)	(1.89)	(2.10)	(2.01)	(2.10)	(2.05)	(0.52)
Panel B: Value-Weighted										
0.58	0.78	0.96	0.85	0.82	0.77	1.05	0.76	1.00	1.26	0.68
(0.80)	(1.05)	(1.35)	(1.23)	(1.18)	(1.14)	(1.68)	(1.27)	(1.63)	(1.94)	(1.72)

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Table: Descriptive Statistics of ROE(TTM) Portfolios

	Low	2	3	4	5	6	7	8	9	High
ROE(TTM)	−11.6%	0.4%	3.3%	5.5%	7.5%	9.6%	11.9%	14.9%	19.3%	30.0%
MCap	48.45	53.97	52.94	60.27	63.38	77.96	100.03	164.87	236.50	300.31
β	1.17	1.18	1.18	1.17	1.16	1.15	1.13	1.11	1.08	1.06
Volatility	48.6%	47.3%	46.6%	46.2%	45.8%	45.4%	44.6%	43.9%	43.3%	43.5%
P/B	7.01	3.86	3.58	3.59	3.70	3.85	4.00	4.21	4.73	5.87

ROE(TTM) is positively correlated with market capitalization. Therefore, the results of the univariate sorting are affected by the size factor. To eliminate this influence, a bivariate sorting is considered.

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Double Sorting

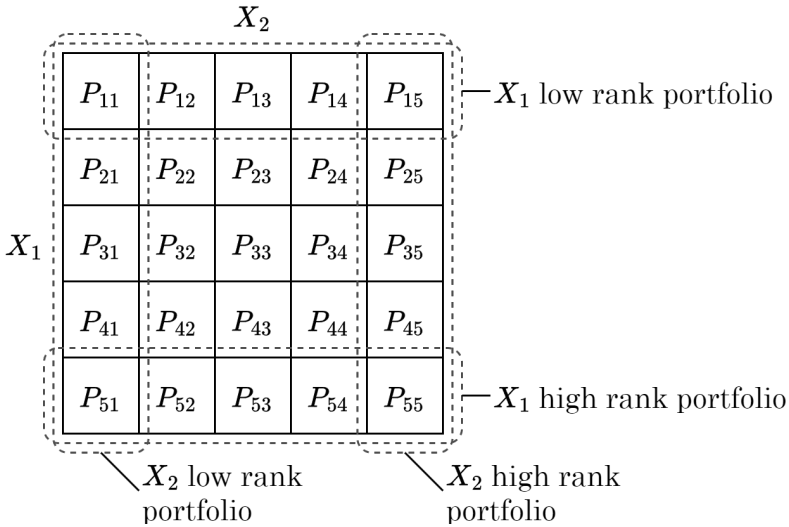
1 Portfolio Sort

- Double sorting involves sorting and constructing factor-mimicking portfolios based on two variables.
- Consider variables X_1 and X_2, where the stocks are divided into L_1 and L_2 groups based on these two variables, resulting in a total of $L_1 \times L_2$ combinations.
- One key detail in double sorting is whether the two variables are sorted independently or if one variable depends on the other. The former is called *independent* double sorting (or *unconditional* double sorting), while the latter is referred to as *dependent* double sorting (or *conditional* double sorting).

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Independent Double Sorting

- A common choise for L_1 and L_2 is to use quintiles for each variable.
- The weights of stocks in these 25 portfolios can be equal-weighted or value-weighted. From these 25 portfolios, factor-mimicking portfolios can be constructed for a given variable.



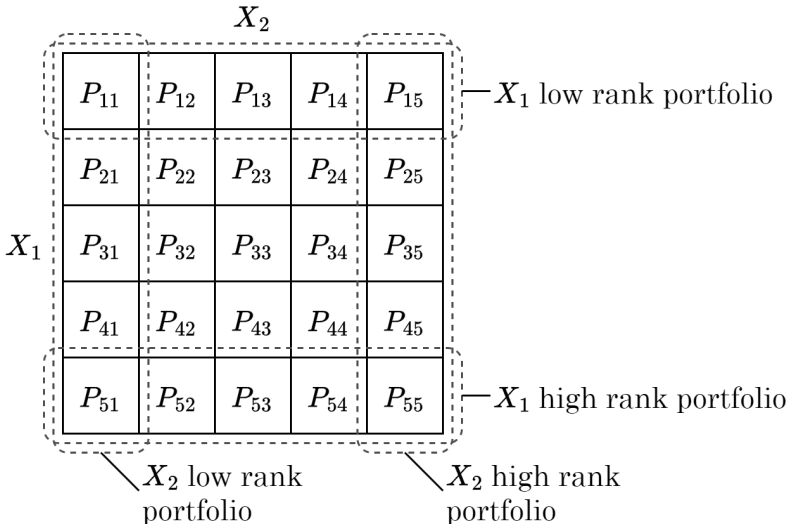
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- The return of the factor $\lambda_{X_1 t}$ is:

$$\lambda_{X_1 t} = \frac{1}{L_2} \sum_{i=1}^{L_2} \left(R_{L_1 i,t} - R_{1i,t} \right).$$

- The return of the factor $\lambda_{X_2 t}$ is:

$$\lambda_{X_2 t} = \frac{1}{L_1} \sum_{i=1}^{L_1} \left(R_{iL_2,t} - R_{i1,t} \right).$$



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Table: ROE(TTM) and MCap Double Sort (Value-Weighted)

	Low	2	3	4	High	High−Low
Small	1.89 (2.34)	2.03 (2.58)	2.17 (2.85)	2.13 (2.88)	2.11 (2.73)	0.22 (1.01)
2	1.16 (1.49)	1.27 (1.71)	1.52 (2.06)	1.83 (2.58)	1.68 (2.27)	0.52 (2.37)
3	0.80 (1.05)	1.06 (1.35)	1.14 (1.57)	1.49 (2.10)	1.42 (2.01)	0.62 (2.84)
4	0.43 (.59)	0.82 (1.11)	0.89 (1.25)	1.18 (1.77)	1.32 (1.95)	0.89 (4.05)
Large	0.60 (0.79)	0.55 (0.78)	0.55 (0.81)	0.74 (1.12)	1.21 (1.83)	0.61 (1.84)
Average	0.98 (1.30)	1.15 (1.56)	1.26 (1.77)	1.48 (2.17)	1.55 (2.24)	0.57 (3.23)

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Potential Issue

- Independent double sorting can lead to certain portfolios containing too few stocks.
- If the *cross-sectional* correlation between X_1 and X_2 is high, a stock with a high X_1 value is also likely to have a high X_2 value. This would cause the groups where both variables are similar—those along the diagonal the above figure —to have more stocks, while other groups (particularly high X_1 but low X_2, and low X_1 but high X_2 groups) would contain fewer stocks.
- This imbalance in the number of stocks within groups can make the factor returns more susceptible to outliers.

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Dependent Double Sorting

- The two variables are applied in a **specific order** for sorting the stocks.
- Suppose we first sort the stocks based on X_1, dividing all stocks into L_1 groups. Then, *within* each of these L_1 groups, we further sort the stocks based on X_2, dividing the stocks into L_2 groups. Again, a total of $L_1 \times L_2$ groups.
- Dependent double sorting focuses on whether the second variable provides **incremental** information in explaining returns after the first variable is controlled for.
- The two sorting variables are **NOT** treated equally: the first sorting variable acts solely as a control variable, and the main interest is the relationship between the second sorting variable and asset returns. **A factor should only be constructed based on the second sorting variable.**

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Fama-MacBeth method is a two-pass estimator.

- The 1st step estimates the factor exposures $\hat{\boldsymbol{\beta}}_i$ for each asset i from N *time-series* regressions:

$$R_{it}^e = a_i + \boldsymbol{\beta}_i^\top \boldsymbol{f}_t + e_{it}, \quad t = 1, 2, \cdots, T, \forall i.$$

- In the 2nd step, we regress the excess return R_{it}^e for period t on the factor exposure estimated at $t-1$, resulting in T cross-sectional regressions:

$$R_{it}^e = \hat{\boldsymbol{\beta}}_i^\top \boldsymbol{\lambda}_t + \alpha_{it}, \quad i = 1, 2, \ldots, N.$$

  This generates a time series of factor returns $\{\hat{\boldsymbol{\lambda}}_t\}$.

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- Estimates:

$$\hat{\boldsymbol{\lambda}} = \frac{1}{T}\sum_{t=1}^{T}\hat{\boldsymbol{\lambda}}_t, \quad \hat{\alpha}_i = \frac{1}{T}\sum_{t=1}^{T}\hat{\alpha}_{it}.$$

- Standard errors:

$$\sigma(\hat{\lambda}_k) = \left[\frac{1}{T^2}\sum_{t=1}^{T}\left(\hat{\lambda}_{kt} - \hat{\lambda}_k\right)^2\right]^{1/2}, \quad \sigma(\hat{\alpha}_i) = \left[\frac{1}{T^2}\sum_{t=1}^{T}\left(\hat{\alpha}_{it} - \hat{\alpha}_i\right)^2\right]^{1/2}.$$

Note that the coefficient above is $\frac{1}{T^2}$ rather than $\frac{1}{T}$ because we are finding standard errors of sample means.

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- Testing pricing errors jointly:

$$\hat{\boldsymbol{\alpha}}^\top \mathrm{cov}(\hat{\boldsymbol{\alpha}})^{-1} \hat{\boldsymbol{\alpha}} \sim \chi^2_{N-K}.$$

where

$$\mathrm{cov}(\hat{\boldsymbol{\alpha}}) = \frac{1}{T^2} \sum_{t=1}^{T} (\hat{\boldsymbol{\alpha}}_t - \hat{\boldsymbol{\alpha}}) (\hat{\boldsymbol{\alpha}}_t - \hat{\boldsymbol{\alpha}})^\top .$$

- Testing factor risk premium:

$$t\text{-statistic} = \frac{\hat{\lambda}_k}{\sigma(\hat{\lambda}_k)}.$$

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- **The advantage of Fama-MacBeth regression is that it can eliminate (or reduce) the impact of cross-sectional correlation of errors on standard error.**

- However, it cannot handle the time-series correlation in α_{it}. But, it is generally believed that the time-series correlation of stock returns is weak.

- In practical applications, it usually assumes that the factor exposures $\boldsymbol{\beta}_i$ may vary across all T periods. In fact, in the original paper, the authors used a rolling window when estimating $\boldsymbol{\beta}_i$, leading to the following cross-sectional regression model:

$$R_{it}^e = \hat{\boldsymbol{\beta}}^\top{}_{it-1}\boldsymbol{\lambda}_t + \alpha_{it}, \quad i = 1, 2, \cdots, N.$$

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Table 2
Fama–MacBeth regressions of stock returns on beta, size, and valuation ratios.

The table reports average slope coefficients from month-by-month Fama–MacBeth regressions. Individual stock returns are regressed cross-sectionally on stock characteristics as of the previous month. The columns correspond to different regression specifications, with nonempty rows indicating the included regressors. The regressors include preranking CAPM β_i estimated using the past 12 months of daily returns with a five-lag Dimson (1979) correction; the log of month-end market cap (logME); the log of book-to-market (logBM); the log of assets-to-market (logAM); EP^+, which equals the positive values of earnings-to-price, and zero otherwise; $D(EP<0)$, which equals one if earnings are negative, and zero otherwise; CP^+; and $D(CP<0)$ (with the last two similarly defined). The last row reports the average adjusted R-squared for each specification. The sample period is January 2000 through December 2016. The t-statistics based on Newey and West (1987) standard errors with four lags are reported in parentheses.

Quantity	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Intercept	0.0149	0.0581	0.0571	0.0659	0.0629	0.0690	0.0564	0.0716	0.0728
	(1.94)	(3.32)	(3.19)	(3.90)	(3.74)	(4.03)	(3.19)	(4.40)	(4.39)
β	−0.0002		−0.0010	−0.0018	−0.0017	0.0002	−0.0010	−0.0002	−0.0004
	(−0.09)		(−0.37)	(−0.71)	(−0.67)	(0.07)	(−0.37)	(−0.06)	(−0.15)
logME		−0.0049	−0.0046	−0.0046	−0.0048	−0.0068	−0.0047	−0.0066	−0.0064
		(−2.91)	(−2.69)	(−2.73)	(−3.00)	(−4.34)	(−2.80)	(−4.49)	(−4.40)
logBM				0.0057				0.0022	0.0035
				(3.21)				(1.31)	(1.76)
logAM					0.0045			0.0014	
					(3.03)			(0.99)	
EP^+						0.9503		0.7825	0.7960
						(4.88)		(4.38)	(5.06)
$D(EP<0)$						0.0006		−0.0005	−0.0001
						(0.31)		(−0.29)	(−0.04)
CP^+							0.0546	0.0181	
							(3.41)	(1.35)	
$D(CP<0)$							0.0019	0.0016	
							(3.11)	(2.37)	
R^2	0.0196	0.0277	0.0441	0.0652	0.0677	0.0615	0.0454	0.0832	0.0776

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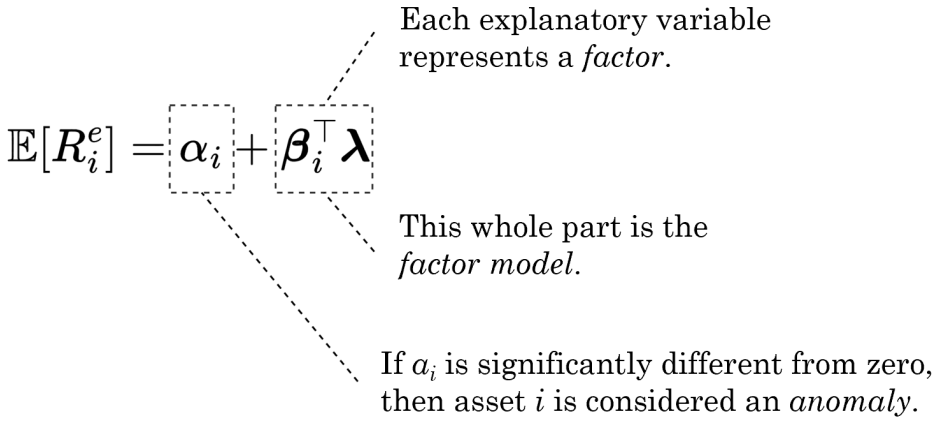


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- Over the past thirty years, scholars have discovered hundreds of anomalies in the U.S. stock market. (But a substantial portion of these findings are false discoveries.)

- An important trend in anomaly research: **using multiple (i.e., composite) variables to construct anomalies.** For example, the *Quality-Minus-Junk* anomaly (it uses numerous variables across three categories).

- This approach has its pros and cons. On the positive side, these anomalies constructed with multiple variables often carry stronger financial implications, which can help people better understand the relationship between firm characteristics and assets expected returns. On the downside, using composite variables can increase the **risk of overfitting**.

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Construct Anomaly Portfolio

Anomalies are typically constructed as portfolios by following these steps:

1. Select a potential financial or technical variable as the anomaly variable.
2. Use portfolio sort to construct the anomaly portfolio using that variable and derive the time series of anomaly returns.
3. Test whether the anomaly generates significant α against a given factor model.

When testing for an anomaly, the null hypothesis is that $\alpha = 0$.

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- To test the null hypothesis, we regress R_t^e on $\boldsymbol{\lambda}_t$:

$$R_t^e = \alpha + \boldsymbol{\beta}_a^\top \boldsymbol{\lambda}_t + \varepsilon_t, \quad t = 1, 2, \cdots, T.$$

- Estimate the model via OLS and use $\hat{\alpha}$ and its standard error s.e.$(\hat{\alpha})$ to construct the t-statistic (d.o.f $= T - K - 1$):

$$t\text{-statistic} = \frac{\hat{\alpha}}{\text{s.e.}(\hat{\alpha})}.$$

- Based on the t-statistic, we determine whether to reject the null hypothesis at a given significance level. If the null hypothesis $\alpha = 0$ is rejected, then the LHS portfolio is considered an anomaly.

Errors may exhibit autocorrelation or heteroscedasticity!

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- Consider the following linear regression model:

$$\begin{aligned} \boldsymbol{y} &= \mathbf{X}\boldsymbol{b} + \boldsymbol{\varepsilon}, \\ \mathbb{E}[\boldsymbol{\varepsilon}|\mathbf{X}] &= \mathbf{0}, \\ \mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^\top|\mathbf{X}] &= \sigma^2\boldsymbol{\Upsilon} = \boldsymbol{\Sigma}. \end{aligned}$$

  The $T \times T$ matrix $\boldsymbol{\Sigma}$ represents the covariance matrix of $\boldsymbol{\varepsilon}$.

- The primary difference between this model and the classical assumptions is the introduction of the positive-definite matrix $\boldsymbol{\Upsilon}$, which allows for autocorrelation and/or heteroscedasticity.

- In classical assumptions, the errors are assumed have a constant variance and are independent, meaning that $\boldsymbol{\Upsilon} = \mathbf{I}$.

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- When heteroscedasticity occurs, $\sigma^2\boldsymbol{\Upsilon}$ takes the following form:

$$\sigma^2\boldsymbol{\Upsilon} = \sigma^2 \begin{bmatrix} \upsilon_{11} & 0 & \cdots & 0 \\ 0 & \upsilon_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \upsilon_{TT} \end{bmatrix} = \begin{bmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_T^2 \end{bmatrix}.$$

- When autocorrelation exists, $\sigma^2\boldsymbol{\Upsilon}$ takes the following form:

$$\sigma^2\boldsymbol{\Upsilon} = \sigma^2 \begin{bmatrix} 1 & \rho_1 & \cdots & \rho_{T-1} \\ \rho_1 & 1 & \cdots & \rho_{T-2} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{T-1} & \rho_{T-2} & \cdots & 1 \end{bmatrix}.$$

- In more general cases, both autocorrelation and heteroscedasticity occur.

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- If the matrix $\boldsymbol{\Upsilon}$ is known, we can use Generalized Least Squares (GLS) estimator. However, when $\boldsymbol{\Upsilon}$ is unknown, OLS is often the preferred method. The OLS estimtor is:

$$\hat{\boldsymbol{b}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{y} = \boldsymbol{b} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon}.$$

- Covariance of the OLS estimator $\hat{\boldsymbol{b}}$, denoted by $\mathbf{V}_{\text{OLS}}$, is:

$$\begin{aligned}
\mathbf{V}_{\text{OLS}} &= \mathbb{E}\left[(\hat{\boldsymbol{b}} - \boldsymbol{b})(\hat{\boldsymbol{b}} - \boldsymbol{b})^\top | \mathbf{X}\right] \\
&= \mathbb{E}\left[(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} | \mathbf{X}\right] \\
&= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\sigma^2 \boldsymbol{\Upsilon}) \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \\
&= \frac{1}{T} \left(\frac{1}{T} \mathbf{X}^\top \mathbf{X}\right)^{-1} \left(\frac{1}{T} \mathbf{X}^\top [\sigma^2 \boldsymbol{\Upsilon}] \mathbf{X}\right) \left(\frac{1}{T} \mathbf{X}^\top \mathbf{X}\right)^{-1}.
\end{aligned}$$

- When $\boldsymbol{\Upsilon} = \mathbf{I}$, $\mathbf{V}_{\text{OLS}} = \sigma^2 \left(\mathbf{X}^\top \mathbf{X}\right)^{-1}$.

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- Covariance of the OLS estimator $\hat{\boldsymbol{b}}$:

$$\mathbf{V}_{\text{OLS}} = \frac{1}{T}\left(\frac{1}{T}\mathbf{X}^\top\mathbf{X}\right)^{-1}\underbrace{\left(\frac{1}{T}\mathbf{X}^\top[\sigma^2\boldsymbol{\Upsilon}]\mathbf{X}\right)}_{\mathbf{Q}}\left(\frac{1}{T}\mathbf{X}^\top\mathbf{X}\right)^{-1}.$$

- The goal is to derive a consistent estimator of $\mathbf{Q}$:

$$\mathbf{Q} \equiv \frac{1}{T}\mathbf{X}^\top[\sigma^2\boldsymbol{\Upsilon}]\mathbf{X} = \frac{1}{T}\sum_{i=1}^{T}\sum_{j=1}^{T}\sigma_{ij}\boldsymbol{x}_i\boldsymbol{x}_j^\top.$$

  where $\boldsymbol{x}_i = [x_{i1}, x_{i2}, \cdots, x_{iK+1}]^\top$, which is the transpose of the i-th row of $\mathbf{X}$.

- Two well-known estimators: White (1980) estimator (adjust for heteroskedasticity) and Newey and West (1987) estimator (adjusts for both heteroskedasticity and autocorrelation).

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- Under heteroscedasticity only, $\mathbf{Q}$ simplifies to:

$$\mathbf{Q} = \frac{1}{T}\sum_{i=1}^{T}\sigma_i^2 \boldsymbol{x}_i\boldsymbol{x}_i^\top.$$

- A consistent estimator of $\mathbf{Q}$ (denoted as $\boldsymbol{S}_0$):

$$\mathbf{S}_0 = \frac{1}{T}\sum_{i=1}^{T}\hat{\varepsilon}_i^2 \boldsymbol{x}_i\boldsymbol{x}_i^\top.$$

- Using $\mathbf{S}_0$, we obtain the estimator for the covariance matrix of $\hat{\boldsymbol{b}}$:

$$\hat{\mathbf{V}}_{\text{OLS}} = T\left(\mathbf{X}^\top\mathbf{X}\right)^{-1}\mathbf{S}_0\left(\boldsymbol{X}^\top\mathbf{X}\right)^{-1}.$$

- Even when the heteroscedasticity's structure or values are unknown, one can still make valid inferences based on the OLS results.

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- In anomaly testing, aside from heteroscedasticity, one must often account for the autocorrelation in $\boldsymbol{\varepsilon}$.
- A natural extension is:

$$\mathbf{S} = \frac{1}{T} \sum_{i=1}^{T} \sum_{j=1}^{T} \hat{\varepsilon}_i \hat{\varepsilon}_j \mathbf{x}_i \mathbf{x}_j^\top.$$

- This seems to be reasonable at the first glance. **BUT...**
  1. It involves a summation of T^2 terms, while the scaling factor is only $1/T$, meaning that $\mathbf{S}$ may not converge.
  2. Even if $\mathbf{S}$ does converge, it may not be positive definite, leading to a non-positive-definite covariance matrix for $\hat{\boldsymbol{b}}$, which is clearly problematic.

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- Newey and West proposed a consistent estimator for $\mathbf{Q}$:

$$\mathbf{S} = \frac{1}{T}\sum_{t=1}^{T}\hat{\varepsilon}_t^2\mathbf{x}_t\mathbf{x}_t^\top + \frac{1}{T}\sum_{j=1}^{J}\sum_{t=j+1}^{T} w_j\hat{\varepsilon}_t\hat{\varepsilon}_{t-j}(\mathbf{x}_t\mathbf{x}_{t-j}^\top + \mathbf{x}_{t-j}\mathbf{x}_t^\top),$$

  where $w_j = 1 - \frac{j}{1+J}$.

- The first term corresponds to the heteroscedasticity-only case, while the second term adjusts for autocorrelation. J is the maximum lag for considering autocorrelation, and w_j is a weight that **diminishes** as lag j increases.

- Newey-West heteroscedasticity and autocorrelation-consistent estimator of the covariance matrix:

$$\hat{\mathbf{V}}_{\text{OLS}} = T(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{S}(\mathbf{X}^\top\mathbf{X})^{-1}.$$

- This estimator is widely used in empirical asset pricing studies.

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Apply Newey-West to Anomaly Testing

1. Perform a time-series regression via OLS to obtain the residuals $\hat{\boldsymbol{\varepsilon}}$.
2. Using $\mathbf{X}$ and $\hat{\boldsymbol{\varepsilon}}$, compute the Newey-West adjusted $\hat{\mathbf{V}}_{\text{OLS}}$. The maximum lag J is determined using the formula from Newey and West (1994):

$$J = \left\lfloor 4 \times \left(\frac{T}{100} \right)^{2/9} \right\rfloor,$$

   where $\lfloor \cdot \rfloor$ represents the floor function, which rounds down to the nearest integer.
3. Take the square root of the diagonal elements of $\hat{\mathbf{V}}_{\text{OLS}}$. These square roots are the standard errors for the regression coefficients $\hat{\boldsymbol{b}}$.
4. Locate the standard error for the intercept, which corresponds to the Newey-West adjusted standard error s.e.$(\hat{\alpha})$ for $\hat{\alpha}$. Use $\hat{\alpha}$ and s.e.$(\hat{\alpha})$ to calculate the t-statistic.

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Table 3

Cross-sectional regressions, 1963–2012.

This table reports the result for four Fama-MacBeth return forecasting regressions. The sample period is from July 1963 to June 2012. The dependent variable is either the focal firm's monthly return *RET* (the first three columns), or the firm's excess return over its value-weighted industry return *RET-INDRET* (Column 4). The explanatory variables include tech-peer return (*TECHRET*), focal firm's value-weighted industry return (*INDRET*), firm size (*SIZE*), book-to-market ratio (*BM*), gross profitability (*GP*), asset growth (*AG*), R&D intensity (*RD*), the firm's own lagged monthly return (RET_{t-1}), and medium-term price momentum (*MOM*). All variables are defined in Appendix Table A1. All explanatory variables are based on the last non-missing available observation for each month t and are assigned to deciles ranging from zero to one. The sample excludes financial firms (one-digit SIC code = 6) and stocks with a price less than \$1 at portfolio formation. Cross-sectional regressions are run every calendar month, and the time-series standard errors are Newey-West adjusted (up to 12 lags) for heteroskedasticity and autocorrelation. Fama-MacBeth *t*-statistics are reported below the coefficient estimates. Coefficients marked with *, **, and *** are significant at 10%, 5%, and 1%, respectively.

Dep. variable ×100	(1) RET_t	(2) RET_t	(3) RET_t	(4) RET_t - $INDRET_t$
$TECHRET_{t-1}$	0.629***	0.583***	0.735***	0.654***
	(4.10)	(5.92)	(6.54)	(6.06)
$INDRET_{t-1}$			0.552***	0.046
			(5.20)	(0.39)

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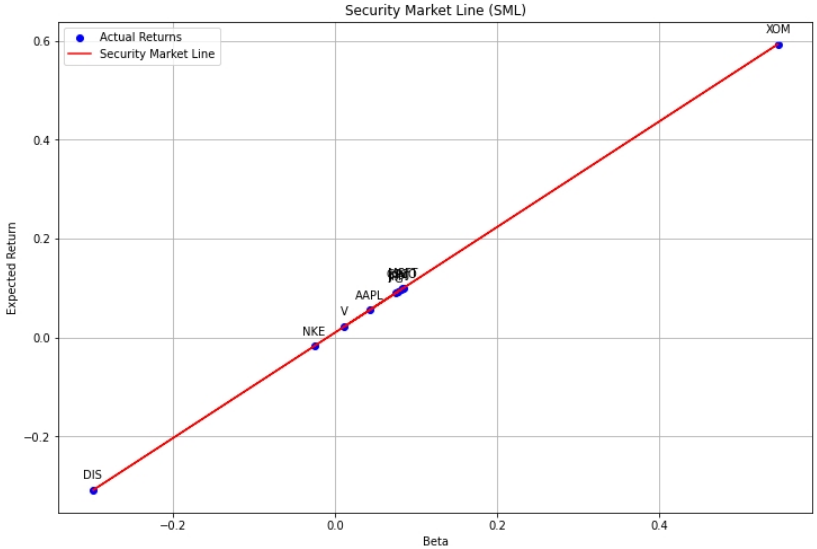
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- The assets used to test a factor model are called *test assets*.
- **Question:** Shall we pick individual assets as test assets? This will lead to *over-rejection* of factor models.
- We can, of course, from an *ex-post* perspective, but that is meaningless. In practice, people often use portfolios as test assets.



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- The primary goal of testing a factor model is to test if the pricing errors of all test assets are **jointly** zero. (For factor risk premiums, construct t-statistics.)

- Recall in Fama-MacBeth regression, we have the following:

$$\hat{\boldsymbol{\alpha}}^\top \text{cov}(\hat{\boldsymbol{\alpha}})^{-1} \hat{\boldsymbol{\alpha}} \sim \chi^2_{N-K}.$$

  The idea is elegant: for the parameters of interest, based on the central limit theorem, they follow an asymptotically multivariate normal distribution and then we construct a quadratic form of these parameters, which yields a test statistic that asymptotically follows a chi-square distribution.

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- GRS did one step further. To test CAPM, they derived a **finite-sample** F distribution for the null hypothesis:

$$\frac{T}{T-K-1}\frac{T-N-K}{N}\left[1+\hat{\theta}^2\right]^{-1}\hat{\boldsymbol{\alpha}}^\top\hat{\boldsymbol{\Sigma}}^{-1}\hat{\boldsymbol{\alpha}}\sim F_{N,T-N-K},$$

where K is the number of factors and $\hat{\theta}^2=\frac{(\bar{f})^2}{\frac{1}{T}\sum_{t=1}^{T}(f_t-\bar{f})^2}$ is the sample squared Sharpe ratio of the market factor.

Compared to the asymptotic chi-square distribution for large samples, this F distribution is the exact distribution for small samples. Therefore, it can provide accurate test results when the sample size (T) is not large.

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- Original GRS paper:

$$\frac{T}{T-K-1}\frac{T-N-K}{N}\left[1+\hat{\theta}^2\right]^{-1}\hat{\boldsymbol{\alpha}}^\top\hat{\boldsymbol{\Sigma}}^{-1}\hat{\boldsymbol{\alpha}}\sim F_{N,T-N-K}.$$

- Cochrane (2005):

$$\frac{T-N-1}{N}\left[1+\hat{\theta}^2\right]^{-1}\hat{\boldsymbol{\alpha}}^\top\hat{\boldsymbol{\Sigma}}^{-1}\hat{\boldsymbol{\alpha}}\sim F_{N,T-N-1}.$$

- Why is that? (I was as confused as you when I first saw the difference!)

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- GRS used OLS estimator of error covariance matrix (unbiased): $\hat{\boldsymbol{\Sigma}}_{\text{OLS}} = \dfrac{1}{T-K-1}\sum_{t=1}^{T}\hat{\boldsymbol{e}}_t\hat{\boldsymbol{e}}_t^\top$.
- Cochrane (2005) used MLE estimator (biased): $\hat{\boldsymbol{\Sigma}}_{\text{MLE}} = \dfrac{1}{T}\sum_{t=1}^{T}\hat{\boldsymbol{e}}_t\hat{\boldsymbol{e}}_t^\top$.
- Two expressions are equivalent:

$$
\begin{aligned}
& \frac{T}{T-K-1}\frac{T-N-1}{N}\left[1+\hat{\theta}^2\right]^{-1}\hat{\boldsymbol{\alpha}}^\top\hat{\boldsymbol{\Sigma}}_{\text{OLS}}^{-1}\hat{\boldsymbol{\alpha}} \sim F_{N,T-N-1} \\
\Rightarrow \quad & \frac{T}{T-K-1}\frac{T-N-1}{N}\left[1+\hat{\theta}^2\right]^{-1}\hat{\boldsymbol{\alpha}}^\top\left[\frac{T-K-1}{T}\hat{\boldsymbol{\Sigma}}_{\text{MLE}}^{-1}\right]\hat{\boldsymbol{\alpha}} \sim F_{N,T-N-1} \\
\Rightarrow \quad & \frac{T-N-1}{N}\left[1+\hat{\theta}^2\right]^{-1}\hat{\boldsymbol{\alpha}}^\top\hat{\boldsymbol{\Sigma}}_{\text{MLE}}^{-1}\hat{\boldsymbol{\alpha}} \sim F_{N,T-N-1}.
\end{aligned}
$$

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- For K factors, then the GRS test statistic becomes (Cochrane 2005):

$$\frac{T-N-K}{T}\left(1+\mathbb{E}_T[\boldsymbol{f}]^\top\hat{\boldsymbol{\Omega}}^{-1}\mathbb{E}_T[\boldsymbol{f}]\right)^{-1}\hat{\boldsymbol{\alpha}}^\top\hat{\boldsymbol{\Sigma}}^{-1}\hat{\boldsymbol{\alpha}}\sim F_{N,T-N-K},$$

where

$$\begin{aligned}\hat{\boldsymbol{\Omega}} &= \frac{1}{T}\sum_{t=1}^{T}\left[\boldsymbol{f}_t-\mathbb{E}_T[\boldsymbol{f}]\right]\left[\boldsymbol{f}_t-\mathbb{E}_T[\boldsymbol{f}]\right]^\top,\\ \hat{\boldsymbol{\Sigma}} &= \frac{1}{T}\sum_{t=1}^{T}\hat{\boldsymbol{e}}_t\hat{\boldsymbol{e}}_t^\top,\end{aligned}$$

and $\mathbb{E}_T[\boldsymbol{f}]$ denotes the sample mean vector of these K factors.

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- **Testing a factor model largely depends on the choice of test assets.**
- When different test assets are used, the test results may reverse, and it becomes difficult to prevent researchers from selecting test assets that best suit their own models (i.e., data snooping).
- In empirical studies, we might encounter situations where adding an extra factor to a factor model leads to an increase in the pricing errors of the test assets, rather than reducing them. (How?)

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- With model B, we have:

$$R = \alpha_R + \beta_1 f_1 + \beta_2 f_2 + \varepsilon, \tag{$\star$1}$$

- With model A, we have:

$$R = \alpha_{R1} + b f_1 + e, \tag{$\star$2}$$

- Assume that factor f_2 and f_1 are related through:

$$f_2 = \alpha_{21} + d f_1 + u. \tag{$\star$3}$$

- Substituting equation ($\star$3) into equation ($\star$1) gives:

$$\begin{aligned} R &= \alpha_R + \beta_1 f_1 + \beta_2(\alpha_{21} + d f_1 + u) + \varepsilon, \\ &= (\alpha_R + \beta_2 \alpha_{21}) + (\beta_1 + \beta_2 d) f_1 + (\beta_2 u + \varepsilon). \end{aligned} \tag{$\star$4}$$

- Finally:

$$\alpha_{R1} = \alpha_R + \beta_2 \alpha_{21}.$$

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- Apply GRS: use factors of one model as *test assets* of the competing model.

	Alphas with respect to:	
Factors	CH-3	FF-3
Panel A: Alpha (t-statistic)		
FFSMB	−0.04	–
	(−0.66)	–
FFHML	0.34	–
	(0.97)	–
SMB	–	0.47
	–	(7.03)
VMG	–	1.39
	–	(7.93)
Panel B: GRS F-statistics (p-value)		
FFSMB, FFHML	0.88	–
	(0.41)	–
SMB, VMG	–	33.90
	–	(2.14×10^{-13})

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- Examine **ex-post** maximum Sharpe ratio spanned by the factors of a given model.

This table reports the maximum annual Sharpe ratio spanned by different factor models, including the composite four-factor model proposed in this paper, the Stambaugh and Yuan (2017) four-factor model, and the Daniel, Hirshleifer and Sun (2020) three-factor model. The dataset ranges from January 1st, 2000 to May 31st, 2021.

model	maximum annual Sharpe ratio
The composite four-factor model	2.02
Stambaugh and Yuan (2017)	0.98
Daniel, Hirshleifer and Sun (2020)	0.80

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- Check number of anomalies against a given factor model.

This table reports numbers of significant test assets given different factor models, including the composite four-factor model proposed in this paper, the Stambaugh and Yuan (2017) four-factor model, and the Daniel, Hirshleifer and Sun (2020) three-factor model. Two t-statistic thresholds for the abnormal return are considered, and they are 2.0 and 3.0. The dataset ranges from January 1st, 2000 to May 31st, 2021.

model	#(t-statistic≥2.0)	#(t-statistic≥3.0)
The composite four-factor model	18	1
Stambaugh and Yuan (2017)	28	7
Daniel, Hirshleifer and Sun (2020)	23	5

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Which Factors?*

Kewei Hou[1], Haitao Mo[2], Chen Xue[3], and Lu Zhang[4]

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Abstract

Many recently proposed, seemingly different factor models are closely related. In spanning tests, the q-factor model largely subsumes the Fama–French five- and six-factor models, and the q^5 model subsumes the Stambaugh–Yuan four-factor model. Their "mispricing" factors are sensitive to the construction procedure, and once replicated via the traditional approach, are close to the q factors, with correlations of

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Misalignment between Academia and Industry

- **Academic Goal:** To propose parsimonious models that **minimize pricing errors** of observed anomalies **in-sample**.
- **Industry Goal:** To **maximize Sharpe ratios** by uncovering complex relationships between a wide range of firm characteristics and future returns **out-of-sample**.
- The traditional limited-factor models, such as the Fama-French Five-Factor model or the Hou-Xue-Zhang model, struggle to continue guiding investment practices effectively.
- For factor investing and quantitative investing, machine learning has the potential to uncover richer insights (as discussed in studies like Gu, Kelly and Xiu 2020; Giglio, Kelly and Xiu 2022). **A new norm has arrived.**

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- Zoo of factors: The covariates related to expected stock returns are high-dimensional:

Data Type	Representative Papers
Trading account	Barber and Odean (2000), Boehmer et al. (2008), Kaniel et al. (2008), Kelley and Tetlock (2013), Barrot et al. (2016)
Order book	Goettler et al. (2009), Cont et al. (2014)
Social media	Antweiler and Frank (2004), Tetlock (2007), Chen et al. (2014), Jame et al. (2016), Bartov et al. (2018), Leung et al. (2019), Umar (2022)
Crowdsourced data	Green et al. (2019), Da et al. (2021)
Text data	Cohen et al. (2020), Bekkerman et al. (2023), Garcia et al. (2023)
Image data	Zhu (2019), Obaid and Pukthuanthong (2022), Jiang et al. (2023)
Weather data	Hirshleifer and Shumway (2003), Goetzmann et al. (2014)
Web search data	Da et al. (2011), Drake et al. (2012), Wu and Brynjolfsson (2015)

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- Nonlinear relationships between covariates and expected returns, e.g., interactions between covariates. (Do you still remember the following example?)

	Highest IVOL	Next 20%	Next 20%	Next 20%	Lowest IVOL	Highest −Lowest	All Stocks
Most Overpriced	−1.89	−0.95	−0.72	−0.47	−0.39	−1.50	−0.81
(Top 20%)	(−12.05)	(−7.39)	(−4.90)	(−3.62)	(−3.04)	(−7.36)	(−8.14)
Next 20%	−0.88	−0.41	−0.31	−0.21	−0.04	−0.84	−0.23
	(−5.86)	(−3.36)	(−3.00)	(−2.08)	(−0.44)	(−4.41)	(−3.88)
Next 20%	−0.09	−0.01	−0.05	−0.12	0.02	−0.10	−0.07
	(−0.53)	(−0.09)	(−0.48)	(−1.29)	(0.18)	(−0.53)	(−1.47)
Next 20%	−0.15	0.07	0.17	0.18	0.23	−0.38	0.18
	(−0.80)	(0.63)	(1.87)	(2.33)	(3.22)	(−1.78)	(4.45)
Most Underpriced	0.56	0.68	0.51	0.33	0.14	0.41	0.28
(Bottom 20%)	(3.27)	(4.91)	(5.02)	(4.10)	(2.04)	(2.16)	(5.67)
Most Overpriced –	−.44	−1.63	−1.23	−0.81	−0.53	−1.91	−1.09
Most Underpriced	(−11.07)	(−8.65)	(−6.43)	(−5.02)	(−3.43)	(−7.62)	(−8.05)
All Stocks	−0.69	−0.12	−0.00	0.05	0.08	−0.78	
	(−6.09)	(−1.56)	(−0.01)	(1.07)	(1.86)	(−5.50)	

Source: Stambaugh, Yu and Yuan (2015)

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6 Econometrics vs. Machine Learning

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- The essential problem in asset pricing and quantitative investment:

$$\begin{aligned} R^e_{i,t+1} &= \mathbb{E}[R^e_{i,t+1}|\boldsymbol{x}_{i,t}] + e_{i,t+1}, \\ \mathbb{E}[R^e_{i,t+1}|\boldsymbol{x}_{i,t}] &= f(\boldsymbol{x}_{i,t}). \end{aligned}$$

- Breiman (2001): *Statistical Modeling: The Two Cultures*:
  - Data modeling culture.
  - Algorithmic modeling culture.

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- **Data modeling culture:** It assumes that the data-generating process is based on a certain probabilistic model, and statistical inference is made based on this assumption. **The main objective of this approach is parameter estimation rather than prediction, aiming to explain causal relationships between variables.** Econometrics belongs to this culture.

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- **Data modeling culture:** It assumes that the data-generating process is based on a certain probabilistic model, and statistical inference is made based on this assumption. **The main objective of this approach is parameter estimation rather than prediction, aiming to explain causal relationships between variables.** Econometrics belongs to this culture.

- **Algorithmic modeling culture: This approach focuses on predictive accuracy rather than model interpretability/parameter estimation.** It does not make strict structural assumptions about the DGP, but instead learns from the data. Machine learning becomes to this culture. The advantage of this approach is its flexibility in handling complex and high-dimensional data without needing to assume the structure in the data.

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- From an econometrics perspective, we approximate f using a linear function of the covariates, $\hat{f}(\boldsymbol{x}_{i,t}) = \boldsymbol{\theta}^\top \boldsymbol{x}_{i,t}$, assuming that $R^e_{i,t+1}$ and $\boldsymbol{x}_{i,t}$ satisfy the following linear regression model:

$$R^e_{i,t+1} = \boldsymbol{\theta}^\top \boldsymbol{x}_{i,t} + e_{i,t+1}.$$

- We can estimate $\boldsymbol{\theta}$ by OLS and conduct statistical inference. Under Gauss-Markov assumptions, OLS estimator is BLUE. Time series regression and Fama-MacBeth regression all follow this culture.

- But what if we care more about predictive accuracy than the unbiasedness of the parameter estimates? (E.g., In shrinkage estimator, we sacrifice unbiasedness to ensure lower variance, thereby reducing overall error!)

- What happens when the number of covariates is large, approaching or even exceeding the number of observations?

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- We do not impose any structural assumptions on the data but rather select a class of models (e.g., neural networks) $\hat{f} \in \mathcal{F}$, and learn the parameters of the model from the data under a given loss function L. View supervised learning as a *function approximation problem*.
- Let $(\boldsymbol{x}_i, y_i)$ represent the i-th observation in the training set. Machine learning estimates the parameters of $\hat{f}$ by minimizing the average loss function across all observations:

$$\min \frac{1}{n} \sum_{i=1}^{n} L(\hat{f}(\boldsymbol{x}_i), y_i).$$

- The core of machine learning is to minimize **generalization error**. To prevent from overfitting the training data, an important concept in machine learning is **regularization**:

$$\min \frac{1}{n} \sum_{i=1}^{n} L(\hat{f}(\boldsymbol{x}_i), y_i) + S(\hat{f}).$$

  The regularization term $S(\hat{f})$ controls the complexity of the model.

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- Let $(\boldsymbol{x}, y)$ represent a new out-of-sample observation, where y is determined by the true model and noise, i.e., $y = f(\boldsymbol{x}) + e$ (assuming the noise e has variance σ^2). On the other hand, the model $\hat{f}$ predicts the value $\hat{f}(\boldsymbol{x})$.

- The generalization error $\mathbb{E}[(y - \hat{f}(\boldsymbol{x}))^2]$ can be decomposed as follows:

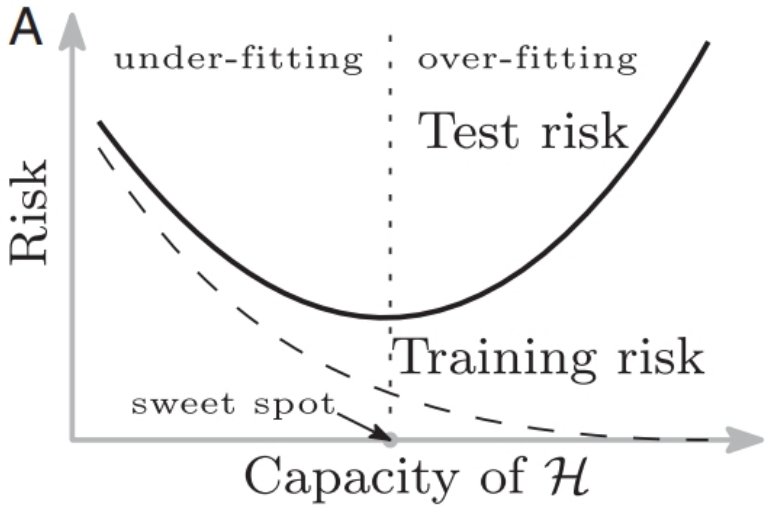
$$\mathbb{E}[(y - \hat{f(\boldsymbol{x})})^2] = \sigma^2 + \text{bias}[\hat{f}(\boldsymbol{x})]^2 + \text{var}[\hat{f}(\boldsymbol{x})].$$

  — **Bias measures the difference between the average prediction of the model and the true value.**
  High bias→ model too simple→ fails to capture patterns in the data (underfitting).

  — **Variance measures the variability of the model's predictions.**
  High variance→ model too complex→ captures noise in the data (overfitting).

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- The optimal model should be flexible enough to capture the inherent relationships in the data, but not so flexible that it models the noise.
- Apply cross-validation to evaluate the generalization ability of the model under different regularization strengths.



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Mullainathan and Spiess (2016)

Estimation vs Prediction

Estimation	Prediction
• Strict assumptions about data generating process	• Allow for flexible functional forms
• Back out parameters	• Get predictions
• Low dimensional	• Does not adjudicate between **observably** similar functions (variables)
$\hat{\beta}$	$\hat{y}_i$

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7 A General Framework

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- Research in asset pricing in the new era:
  - **What it is**: Based on big data, utilizing machine learning algorithms as tools, and centered around asset pricing theory.
  - **What it is not**: Blindly feeding data into complex algorithms, expecting data to speak for itself.

Placing asset pricing within a theoretical framework to explore cutting-edge advancements helps to understand the commonalities and differences between various methods, thus bridging the latest academic findings with practical investment applications.

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- Without loss of generality, we can write:

$$m_{t+1} = 1 - \boldsymbol{b}_t^\top \boldsymbol{R}_{t+1}^e.$$

- View weights $\boldsymbol{b}_t$ as a function of firm characteristics:

$$m_{t+1} = 1 - \boldsymbol{b}_t^\top \boldsymbol{R}_{t+1}^e, \quad \text{with } b_{i,t} = g(\boldsymbol{c}_{i,t}),$$

where $\boldsymbol{c}_{i,t}^\top$ represents the ith row of the $N \times L$ matrix $\mathbf{C}_t$, and $g(.)$ is a (nonlinear) function. $\mathbf{C}_t$ summarizes all relevent firm characteristics of N assets at time t. The information set can be high-dimensional (i.e., L can be large).

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- Approximate $g(.)$ by a set of *managed portfolios*:

$$g(\boldsymbol{c}_{i,t}) \approx \sum_{k=1}^{K} \theta_k g_k(\boldsymbol{c}_{i,t}).$$

- Convert pricing model to *unconditional* model:

$$m_{t+1} = 1 - \sum_{k=1}^{K} \theta_k \tilde{R}^e_{k,t+1}, \quad \text{with } \tilde{R}^e_{k,t+1} = \sum_{i=1}^{N} g_k(\boldsymbol{c}_{i,t}) R^e_{i,t+1}.$$

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- The Framework:

$$m_{t+1} = 1 - \sum_{k=1}^{K} \theta_k \tilde{R}^e_{k,t+1}, \quad \text{with } \tilde{R}^e_{k,t+1} = \sum_{i=1}^{N} g_k(\boldsymbol{c}_{i,t}) R^e_{i,t+1}.$$

- Different pricing models represent different **priors** regarding the factors embedded in the true SDF. For example, in the Fama and French (1993) three-factor model, the firm characteristics are market cap and book-to-market ratio, and the function g_k is a double portfolio sort.

- Research objective: The goal is to **maximize the out-of-sample Sharpe ratio** of the MVE portfolio by selecting firm characteristics $\mathbf{C}$, constructing the managed porfolios g_k, and estimating the SDF coefficients $\boldsymbol{\theta} = [\theta_1, \cdots, \theta_k]^\top$.

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$$m_{t+1} = 1 - \sum_{k=1}^{K} \theta_k \tilde{R}^e_{k,t+1}, \quad \text{with } \tilde{R}^e_{k,t+1} = \sum_{i=1}^{N} g_k(\boldsymbol{c}_{i,t}) R^e_{i,t+1}.$$

- Research about $\mathbf{C}$:
  - Which firm characteristics are related to expected returns? (Luck vs. skill)
  - In the high-dimensional era, when the rational expectation assumption fails, are there explanations beyond risk compensation and mispricing behind in-sample predictability?
  - Can economic principles guide us in finding better firm characteristics?
- Research about $g_k()$:
  - What is the nature of the nonlinear relationship between firm characteristics and expected returns (e.g., higher orders, interactions)?

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- Multiple hypothesis testing (Harvey, Liu and Zhu 2016; Harvey 2017; Harvey and Liu 2020; Harvey, Sancetta and Zhao 2025).
- Illusion of sparsity and the problem of high-dimensional learning by investors (Martin and Nagel 2022).
- Guidance from APT: Covariates that explain expected returns should also explain the comovement of assets (Kozak, Nagel and Santosh 2018).

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- Suppose there are K covariates (i.e., factors) being tested, with K_0 true null hypotheses and K_1 false null hypotheses. At a given significance level (e.g., 0.05), L null hypotheses are rejected, including T_1 true discoveries and F_1 false discoveries.

	$\mathcal{H}_0$ Accepted	$\mathcal{H}_0$ Rejected	Total
$\mathcal{H}_0$ is True	T_0	F_1	K_0
$\mathcal{H}_0$ is False	F_0	T_1	K_1
Total	$K - L$	$L = F_1 + T_1$	K

- Family-wise error rate: $\text{prob}(F_1 \geq 1) \leq \alpha$.
- False discovery rate: $\mathbb{E}[F_1/L] \leq \alpha$.
- False discovery proportion: $\text{prob}(F_1/L \geq \gamma) \leq \alpha$.

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Frequentist

Family-wise error rate

- Bonferroni (1936)
- Holm (1979)
- White (2000)
- Romano and Wolf (2005, 2007)

False discovery rate

- Benjamini and Hochberg (1995)
- Benjamini and Yekutieli (2001)
- Harvey, Liu and Zhu (2016)

False discovery proportion

- Romano and Wolf (2007)
- Romano, Shaikh and Wolf (2008)
- Chordia, Goyal and Saretto (2020)

Bayesian

Full Bayesian

- Scott and Berger (2006)
- Harvey, Liu and Zhu (2016)

Minimum Bayesian factor

- Harvey (2017)

Bayesian extension

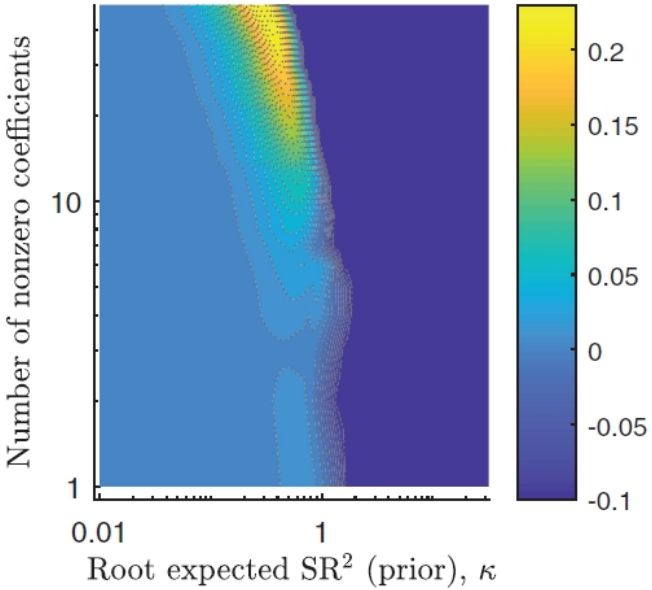
- Harvey and Liu (2020, 2021)

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- Giannone, Lenza and Primiceri (2021) studied six issues in the field of economics/finance in the era of big data under the title *Illusion of Sparsity*, one of which is cross-sectional return prediction:
  - **The probability of inclusion is positively correlated with the degree of shrinkage.**
  - **The assumption of sparsity does not hold, as many covariates are related to predictive returns.**
  - **There is significant model uncertainty, and covariates are correlated with each other.**
- Implications for out-of-sample prediction: sparsity + ignoring uncertainty will lead to a loss of predictability.

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- The horizontal axis represents prior strength, while the vertical axis shows the number of firm characteristics used.
- The optimal out-of-sample predictability requires both:
  - Prior constraints.
  - Almost all characteristics contributing to the construction of the SDF, indicating that sparsity does not hold.
- It is difficult to construct the SDF based on a limited number of firm characteristics.



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- Rational expectation assumes that investors know the true valuation model. However, investors face a high-dimensional learning problem, making this impossible.
- In equilibrium, asset prices deviate from what would be expected under rational expectations; in **ex post** analysis, realized returns contain a predictable component **due to estimation errors**.

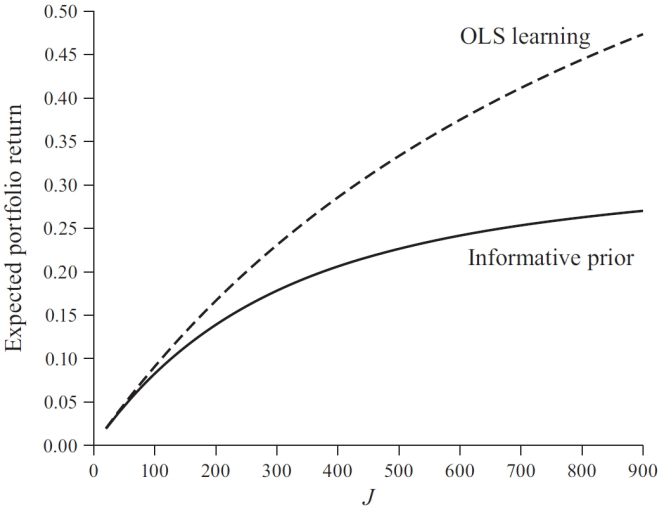
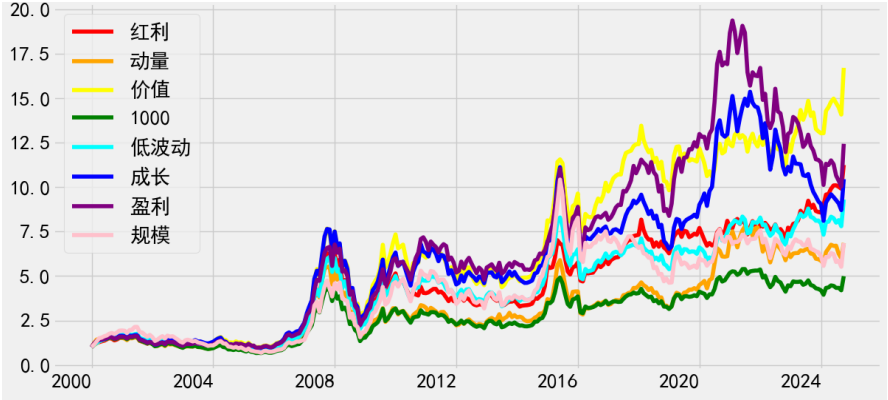


Figure 5.2. Adjusted R^2 in in-sample return prediction regression

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Factor	Mean	Std	SR
Value	1.27	7.92	0.55
Profitability	1.16	7.84	0.51
Dividend	1.09	7.34	0.51
Low Vol	1.04	7.57	0.48
Size	1.06	9.08	0.40
Momentum	1.00	8.28	0.42
Growth	1.14	8.36	0.47

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- Under the assumption that there are no risk-free arbitrage opportunities in the market (i.e., no investment opportunities with extremely high Sharpe ratios), the factors with high risk premiums must also be the primary sources of stock comovement (Kozak, Nagel and Santosh 2018).

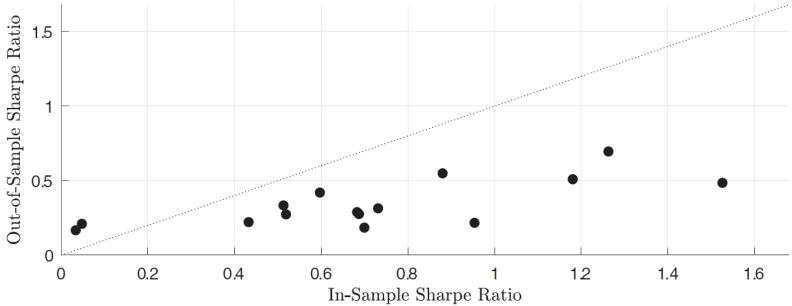


Figure 4. In-sample and out-of-sample Sharpe ratios. The figure shows in-sample and out-of-sample Sharpe ratios of 15 anomaly long-short strategies. The sample period is split into two halves. In-sample Sharpe ratios are those in the first subperiod. Out-of-sample Sharpe ratios are those in the second subperiod. Sharpe ratios are annualized.

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Find $g_k()$: No-Free-Lunch

- Wolpert (1996): No algorithm is inherently better than another.

Machine Learning		Ad-hoc Models	
Model	OOS Sharpe	Model	OOS Sharpe
BPZ Random Forest	2.19	FF3	0.45
Davis Neural Network	3.21	FF6	1.03
KNS PCA	3.21	HXZ	1.10
KPS IPCA	3.39	SY	0.61

Baba-Yara, Boyer and Davis (2021)

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Off-the-Shelf Solutions are Hard to Succeed

- Parameter scaling, regularization, cost function... every choice can affect the out-of-sample risk-return characteristics.

Return prediction with a polynomial of lagged returns

						CV portfolio return r_p		
Method	Scaling	CV criterion	γ (i)	IS R^2 (ii)	CV R^2 (iii)	Mean (iv)	S.D. (v)	Sharpe Ratio (vi)
OLS	Equal	n/a	0	5.22	−1.18	4.12	11.60	0.35
Ridge	Equal	R^2	2.25	2.63	0.84	4.20	13.85	0.30
Ridge	Unequal	R^2	1.40	2.69	1.18	4.55	12.47	0.37
Ridge	Unequal	$E[r_p]$	3.11	1.75	0.89	4.58	12.94	0.35
Lasso	Unequal	R^2	0.00028	3.55	0.84	4.25	11.79	0.36

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- Example: Empirical analysis shows that ridge regression outperforms OLS in terms of out-of-sample R^2 and Sharpe Ratio. Ridge regression has an elegant *Bayesian shrinkage interpretation* behind it.
- Posterior estimate of the parameter $\boldsymbol{\theta}$:

$$\hat{\boldsymbol{\theta}} = \left(\mathbf{X}^\top \mathbf{X} + \frac{\sigma^2}{\sigma_g^2}\mathbf{I}\right)^{-1} \mathbf{X}^\top \boldsymbol{y}.$$

- Ridge regression:

$$\hat{\boldsymbol{\theta}} = \left(\mathbf{X}^\top \mathbf{X} + \gamma\mathbf{I}\right)^{-1} \mathbf{X}^\top \boldsymbol{y}.$$

- Overfitting: placing too much emphasis on data while ignoring prior experience. **Ignoring prior experience in investment is dangerous.**

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	Value					Quality				
	Growth	Q2	Q3	Q4	Value	Low	Q2	Q3	Q4	High
Big	−5.13%	−3.60%	2.70%	5.93%	7.87%	−4.44%	−0.06%	0.17%	5.51%	5.87%
Q2	−4.16%	−1.68%	3.02%	7.04%	8.26%	−2.86%	1.39%	1.86%	1.63%	9.42%
Q3	−6.65%	−3.54%	3.69%	4.53%	8.63%	−2.23%	−2.60%	1.09%	2.79%	9.09%
Q4	−5.83%	−3.41%	−0.36%	2.11%	5.34%	−5.33%	−2.28%	0.15%	−0.10%	5.65%
Small	−7.34%	−2.06%	0.26%	3.27%	7.75%	−4.28%	−1.23%	−0.55%	2.47%	4.80%

	Momentum					Volatility				
	Loser	Q2	Q3	Q4	Winner	High	Q2	Q3	Q4	Low
Big	−3.28%	1.31%	2.51%	3.47%	4.62%	−9.03%	−7.07%	0.74%	−3.01%	6.23%
Q2	2.08%	3.59%	6.04%	2.40%	0.61%	−5.78%	−5.64%	0.85%	6.98%	4.49%
Q3	−0.88%	1.82%	3.35%	0.29%	2.83%	−0.90%	1.50%	−1.49%	3.78%	8.19%
Q4	−0.88%	−0.01%	−1.63%	1.28%	0.04%	−2.14%	0.72%	0.96%	4.93%	5.54%
Small	−1.45%	1.07%	3.31%	−0.28%	0.36%	1.06%	−.66%	−0.11%	4.76%	3.70%

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	Value					Quality				
	Growth	Q2	Q3	Q4	Value	Low	Q2	Q3	Q4	High
Big	14.75%	−1.30%	−0.61%	−1.91%	−5.20%	−5.93%	−0.56%	4.60%	2.05%	14.79%
Q2	−6.16%	−6.90%	−7.93%	−1.71%	−10.10%	−12.16%	−5.05%	−7.32%	−6.28%	−2.74%
Q3	−5.67%	−5.47%	−2.82%	−6.24%	−5.30%	−12.33%	−10.47%	−3.58%	−1.71%	3.65%
Q4	−8.38%	−6.60%	−10.64%	−7.17%	−10.76%	−12.56%	−13.06%	−10.34%	−5.86%	−3.01%
Small	−5.97%	−4.13%	−4.62%	−6.18%	−8.09%	−11.43%	−7.77%	−6.01%	−2.59%	−1.85%

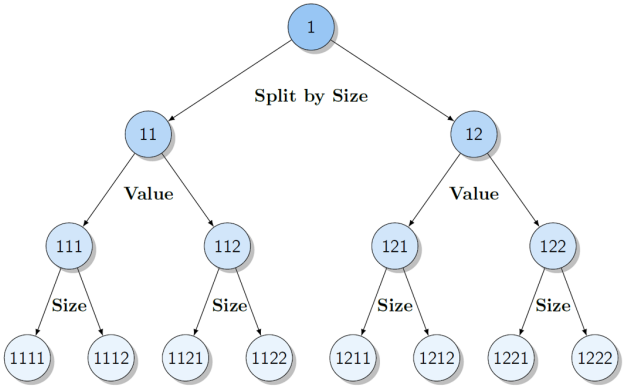
	Momentum					Volatility				
	Loser	Q2	Q3	Q4	Winner	High	Q2	Q3	Q4	Low
Big	−5.06%	−2.28%	−1.61%	11.89%	12.71%	1.12%	−9.63%	−11.93%	−16.64%	−11.28%
Q2	−10.26%	−4.46%	−6.39%	−0.77%	−7.86%	11.77%	−4.92%	−6.27%	2.59%	−19.87%
Q3	−8.09%	−7.54%	−2.50%	−7.48%	−0.41%	16.00%	10.38%	−2.34%	−9.39%	−8.54%
Q4	−9.75%	−11.90%	−14.85%	−3.39%	−4.70%	4.46%	11.16%	−5.62%	−0.98%	−8.79%
Small	−10.14%	−6.86%	−5.23%	−4.57%	−1.34%	14.78%	12.33%	−1.71%	−8.03%	−7.34%

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Example: Bryzgalova, Pelger and Zhu (2020)

$$m_{t+1} = 1 - \sum_{k=1}^{K} \theta_k \tilde{R}^e_{k,t+1}, \quad \text{with } \tilde{R}^e_{k,t+1} = \sum_{i=1}^{N} g_k(\boldsymbol{c}_{i,t}) R^e_{i,t+1}.$$

- Construct $g_k()$ by decision tree.
- When estimating weights θ_k, simultaneously shrink both the expected returns and the covariance matrix of the factors.

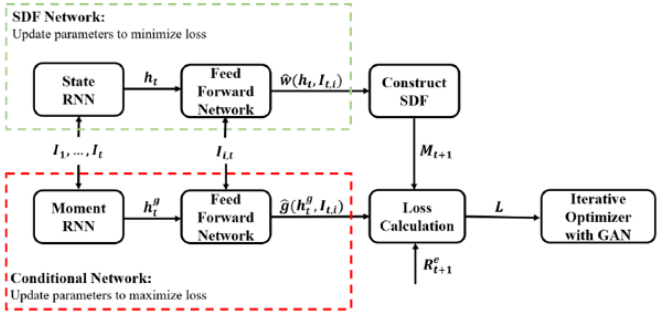


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Example: Chen, Pelger and Zhu (2020)

$$m_{t+1} = 1 - \sum_{k=1}^{K} \theta_k \tilde{R}^e_{k,t+1}, \quad \text{with } \tilde{R}^e_{k,t+1} = \sum_{i=1}^{N} g_k(\boldsymbol{c}_{i,t}) R^e_{i,t+1}.$$

- Generative Adversarial Network (GAN).
- Given $f(.)$, the SDF network estimates parameters with the goal of minimizing pricing errors.
- Given the SDF, the conditional network selects new test assets with the goal of maximizing pricing errors.



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$$R^e_{i,t+1} = \boldsymbol{\beta}_{i,t}(\boldsymbol{c}_t)^\top \boldsymbol{F}_{t+1} + \epsilon_{i,t+1}.$$

- Model $\boldsymbol{\beta}$ by considering it as a nonlinear function of firm characteristics (and other potential variables, such as macroeconomic variables).
- $\boldsymbol{\beta}$ maps a large number of firm variables related to expected returns into a limited number of latent factors.
- Once $\boldsymbol{\beta}$ is obtained, estimating factor premiums is straightforward, i.e., $\boldsymbol{F}_{t+1} = (\boldsymbol{\beta}_t^\top \boldsymbol{\beta}_t)^{-1}\boldsymbol{\beta}_t^\top \boldsymbol{R}^e_{t+1}$.

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- Instrumented PCA:

$$\underbrace{\boldsymbol{\beta}_{i,t}}_{K\times 1} = \underbrace{\boldsymbol{\Gamma}}_{K\times P}\underbrace{\boldsymbol{c}_{i,t}}_{P\times 1}.$$

- P is the number of firm characteristics, K is the number of factors ($K \ll P$), and the matrix $\boldsymbol{\Gamma}$ reduces the high-dimensional firm characteristics to K latent factors through K different linear combinations.

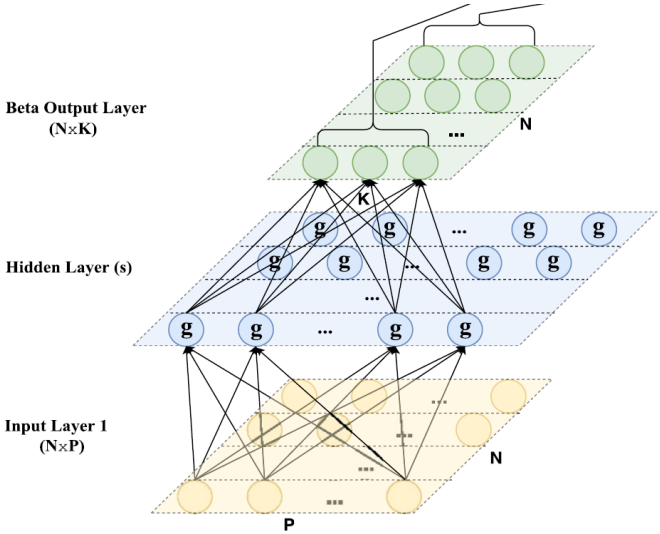
- Estimate $\boldsymbol{\Gamma}$ by minimizing pricing errors.

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- Autoencoder:

$$\boldsymbol{\beta}_{i,t} = f(\boldsymbol{c}_{i,t}) \quad f : \mathbb{R}^P \to \mathbb{R}^K.$$

- Mitigate overfitting by regularization.
- Estimate model parameters by minimizing pricing errors.
- When the activation function is linear, the autoencoder is similar to PCA (we will discuss the differences in a future lecture).



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Questions?

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